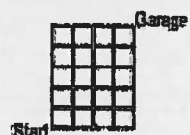
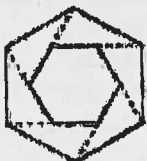


Tour de Maths 2014 Senior Leg 2 - SJC

Question Paper

1 (4 marks) The digits 1, 2, 3, 4, and 9 are each used once to form the smallest possible even five-digit number. The digit in the tens place is _____. (A) 1 (B) 2 (C) 3 (D) 4 (E) 9	2 (4 marks) If $a \oplus b = \frac{a+b}{a-b},$ then $(6 \oplus 4) \oplus 3 = \underline{\hspace{2cm}}$ (A) 4 (B) 13 (C) 15 (D) 30 (E) 72	3 (4 marks) On a twenty-question test, each correct answer is worth 5 points, each unanswered question is worth 1 point, and each incorrect answer is worth 0 points. Which of the following scores is not possible? (A) 80 (B) 91 (C) 92 (D) 95 (E) 97
4 (4 marks) The ratio $\frac{10^{2000} + 10^{2002}}{10^{2001} + 10^{2001}}$ is closest to which of the following numbers? (A) 0.1 (B) 0.2 (C) 1 (D) 5 (E) 10	5 (4 marks) Two different positive numbers a and b each differ from their reciprocals by 1. What is $a + b$? (A) 1 (B) 2 (C) $\sqrt{5}$ (D) $\sqrt{6}$ (E) 3	6 (4 marks) $A, B, C, D,$ and E represent five different digits. What is $B + D$? $\frac{A}{+B} \quad \frac{C}{+D}$ $\frac{C}{EA}$
7 (5 marks) A box contained 31 chocolates. The first day, Gabriela ate $\frac{3}{4}$ of the number that Pam ate. The second day, Gabriela ate $\frac{2}{3}$ of the number that Pam ate that day, and the chocolates were all gone. How many chocolates did Gabriela eat?	8 (5 marks) Alisha, Brian, Cassie, Dianne, Earl, and Fred form a circle. Each picks a number and tells it to the two neighbors on each side. They announce the average of the numbers of their immediate neighbors, as given in the figure.  What number did Dianne pick?	9 (5 marks) A painted $2 \times 2 \times 2$ cube is cut into eight unit cubes. What fraction of the total surface area of the eight small cubes is painted?
10 (5 marks) Three angles of a pentagon have measures 88°, 124°, and 92°. If the measures of the remaining two angles are equal, what is the measure, in degrees, of one of the remaining angles?	11 (5 marks) Three solid balls, each of radius 3.25 cm, are stored in a cylindrical can with the smallest possible radius and volume. What fraction of the can's volume is air?	12 (5 marks) If $\sqrt{19^2 + 19^2 + 19^2 + \dots + 19^2} = 19^2$ how many times must 19^2 appear in the radicand?
13 (5 marks) A transit bus can travel in only two directions, north and east. From its starting point on the map shown, determine how many paths exist to reach the garage. 	14 (5 marks) Jerry, Kate, and Kelsey climb a flight of 102 steps. Kelsey climbs 1 at a time, Jerry climbs 2 at a time, and Kate climbs 3 at a time. What is the total number of steps used by exactly two people?	15 (5 marks) Find the smallest number greater than 2000 that is divisible by 2, 3, 4, 5, and 6.

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16 (5 marks) Al, Ed, and Tom are different ages. Exactly one of the following three statements is true: I. Ed is the oldest. II. Al is not the oldest. III. Tom is not the youngest. Who is the youngest boy?	17 (5 marks) The ages of Moe and Joe now total 49 years. Moe's age now is twice the age that Joe was when Moe's age was the same as Joe's is now. How old are Moe and Joe now?	18 (5 marks) If 100 is divided by the positive integer x, the remainder is 2. What is the remainder when 198 is divided by x?
19 (6 marks) A train traveling between two towns arrives 9 minutes late when traveling at 36 KPH. When it travels at 27 KPH, the train arrives 39 minutes late. What is the distance between the two towns?	20 (6 marks) Thirty cards from an ordinary deck of fifty-two playing cards are placed in a pile on a table. What is the difference between the number of black cards in this pile and the number of red cards in the rest of the deck?	21 (6 marks) Before she took her last exam, the average of Tara's exam scores was 89. She figures that if she scores 97 on the last exam, her average for all the exams will be 90. If she scores a 73, her average for all the exams will be 87. How many exams, including the last one, are given in the class?
22 (6 marks) A sandbox in the shape of an equilateral triangle is 10 meters on each side. A fence is built around the triangular sandbox at a constant distance of 4 meters. What is the number of square meters in the area between the fence and the triangular sandbox? Express your answer in terms of π.	23 (6 marks) Each side of a regular hexagon is extended by an amount equal to its length, as shown, and the endpoints of these segments are joined to form a larger regular hexagon. What is the ratio of the area of the larger hexagon to the area of the smaller hexagon? 	24 (6 marks) Define a function by $f(x) = \frac{1}{1-x}$ $x \neq 0, 1$. What is $f(f(f(x)))$?

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Memo

Marks/Quest	1	2	3	TOTALS
4	E	A	E	
	4	5	6	
4	D	C	10	
	7	8	9	
5	13	4	$\frac{1}{2}$	
	10	11	12	
5	118°	$\frac{1}{3}$	361	
	13	14	15	
5	126	51	2040	
	16	17	18	
5	Ed	28 and 21	2	
	19	20	21	
6	54 km	4	8	
	22	23	24	
6	120+16 π	3 : 1 (or 3)	a	
			TEAM TOTAL	

